Estimation of Thin Ice Thickness and Detection of Fast Ice from SSM/I Data in the Antarctic Ocean

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(Manuscript received 1 March 2006, in final form 29 March 2007)

ABSTRACT

Antarctic coastal polynyas are important areas of high sea ice production and dense water formation, and thus their detection including an estimate of thin ice thickness is essential. In this paper, the authors propose an algorithm that estimates thin ice thickness and detects fast ice using Defense Meteorological Satellite Program (DMSP) Special Sensor Microwave Imager (SSM/I) data in the Antarctic Ocean. Detection and estimation of sea ice thicknesses of <0.2 m are based on the SSM/I 85- and 37-GHz polarization ratios (PR_{85} and PR_{37}) through a comparison with sea ice thicknesses estimated from the Advanced Very High Resolution Radiometer (AVHRR) data. The exclusion of data affected by atmospheric water vapor is discussed. Because thin ice and fast ice (specifically ice shelves, glacier tongues, icebergs, and landfast ice) have similar PR signatures, a scheme was developed to separate these two surface types before the application of the thin ice algorithm to coastal polynyas. The probability that the algorithm correctly distinguishes thin ice from thick ice and from fast ice is ~95%, relative to the ice thicknesses estimated from AVHRR. Although the standard deviation of the difference between the thin ice thicknesses estimated from the SSM/I algorithm and AVHRR is ~0.05 m and thus not small, the estimated ice thicknesses from the microwave algorithm appear to have small biases and the accuracies are independent of region and season. A distribution map of thin ice occurrences derived from the SSM/I algorithm represents the Ross Sea coastal polynya being by far the largest among the Antarctic coastal polynyas; the Weddell Sea coastal polynyas are much smaller. Along the coast of East Antarctica, coastal polynyas frequently form on the western side of peninsulas and glacier tongues, downstream of the Antarctic Coastal Current.

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DOI: 10.1175/JTECH2113.1

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1. Introduction

Most Antarctic coastal polynyas are latent heat polynyas, which are formed by the divergent ice motion due to prevailing winds or oceanic currents (Morales Maqueda et al. 2004). Recurring coastal polynyas are found leeward of prevailing offshore winds or leeward of the Antarctic Coastal Current in areas of fast ice such as landfast ice, glacier tongues, and grounded icebergs. For Antarctic coastal polynyas over the continental shelf, the oceanic heat from below is expected to be small because the whole water column likely reaches the freezing point temperature in winter due to its shallowness (Zwally et al. 1985). Therefore, most of the heat loss is balanced by sea ice production, and Antarctic coastal polynyas are considered to be the sites of high sea ice production (Gordon and Comiso 1988). Detection of coastal polynyas by use of satellite data and the subsequent estimation of sea ice production through the calculation of surface heat fluxes has proven to be an effective way to estimate ice production in these areas. Some regional studies have been undertaken along the Antarctic Wilkes Land coast (e.g., Cavalieri and Martin 1985), and along the Weddell Sea coast (e.g., Markus et al. 1998; Renfrew et al. 2002). Satellite microwave radiometer data are available for the whole Antarctic Ocean every day regardless of darkness or cloud cover. These data provide a means to continuously monitor coastal polynyas.

Under winter conditions, open water in Antarctic coastal polynyas often freezes very rapidly. Thus, most of the polynya area is covered with thin ice, except possibly within 1 km from the coastline in winter (Pease 1987). For instance, in the Ross Sea coastal polynya, ice thicknesses reach up to ~0.2 m (Jeffries and Adolphs 1997). Usually the thin ice region extends to no more than 100 km from the coastline (Smith et al. 1990). The present study defines the combined open water area and thin ice region as a coastal polynya. With the microwave radiometer data whose spatial resolution is between 12.5 and 50 km, depending on frequency, the open water in a coastal polynya is usually too small to be resolved; hence, coastal polynyas are identified as areas of thin ice. In winter, heat loss over thin ice regions is very large and sensitive to ice thickness. Thus, to discuss heat loss and sea ice production in coastal polynyas, it is necessary to map the thin ice region and to estimate the thin ice thickness as accurately as possible.

The National Aeronautics and Space Administration (NASA) Team algorithm (Cavalieri et al. 1984), the NASA Team 2 (NT2) algorithm (Markus and Cavalieri 2000), and the Bootstrap algorithm (Comiso et al. 1984; Comiso 1986) provide sea ice concentration but not ice type or ice thickness information. When these algorithms are applied to thin ice regions, the retrieved ice concentration often has a low concentration bias. For the detection of thin ice, a modified algorithm is necessary. Cavalieri (1994) provided a thin ice algorithm for the seasonal sea ice zone of the Northern Hemisphere. This algorithm divides sea ice into three categories: new ice, young ice, and first-year ice. Though the algorithm distinguishes thin ice from thick ice, it does not provide specific ice thickness information. Because the algorithm uses 19-GHz data, the spatial resolution is coarse, ~60 km. An algorithm called the Polynya Signature Simulation Method (PSSM; Markus and Burns 1995) detects Antarctic coastal polynyas using the polarization ratio (PR) defined as (TBv – TBh)/(TBv + TBh), where TBv and TBh are the microwave radiances at a given frequency. This algorithm uses 85-GHz data, which have a higher spatial resolution than does the 19-GHz data, and reduces the atmospheric effects through the combination of 85- and 37-GHz data. However, the algorithm classifies areas into only three categories: open water, thin ice of 0.1-m thickness, and first-year ice. Martin et al. (2004) presented an algorithm, which provides specific thicknesses of thin ice <0.1–0.2 m in the Chukchi Sea Alaskan coast polynya. Their proposed equation of ice thickness estimation is based on the ratio of vertically and horizontally polarized 37 GHz in comparison with the ice thickness estimated from the Advanced Very High Resolution Radiometer (AVHRR) data. With a spatial resolution of 25 km, this algorithm is only applicable to relatively large coastal polynyas.

In this study, we mainly use the 85-GHz brightness temperature data of the Defense Meteorological Satellite Program (DMSP) Special Sensor Microwave Imager (SSM/I) with a spatial resolution of 12.5 km. Care must be paid to atmospheric effects when using 85-GHz data. In situ observations of ice thickness in Antarctic coastal polynyas are very limited. At present, we have no ice thickness data on scales comparable to SSM/I grid cells. Surface temperature data from AVHRR with a spatial resolution of 1–4 km possibly can provide ice thickness estimation through heat flux calculations (Yu and Rothrock 1996; Drucker et al. 2003; Tamura et al. 2006). In this study, we use ice thickness information estimated from AVHRR data for comparison with and validation of the SSM/I retrievals. For validation, we choose four polynya areas (Fig. 1): the Weddell Sea coastal polynyas, the Ross Sea coastal polynya, the Mertz Glacier Polynya, and the Cape Darnley Polynya (naming is based on Massom et al. 1998). In the Antarctic Ocean, various types of fast ice and ice shelves

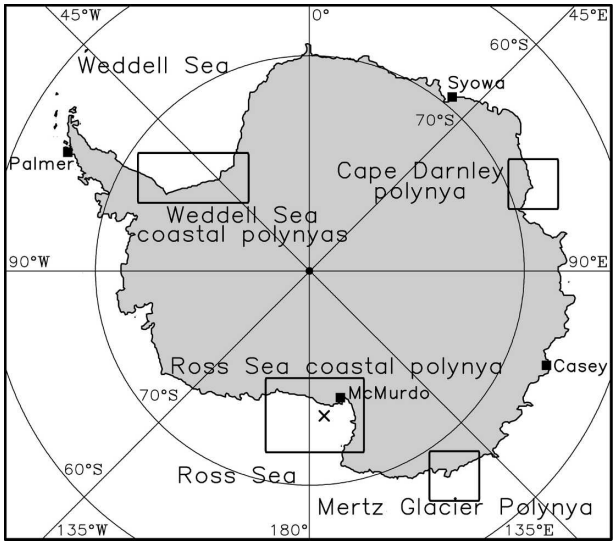

FIG. 1. Map of the Antarctic Ocean. The four rectangular boxes outline the validation areas for algorithm described in this paper. The × symbol indicates the location of the wind data used in Fig. 12.

are found adjacent to coastal polynyas. Distinguishing between these ice features and thin ice is difficult because of their similar microwave characteristics. Detection of fast ice and ice shelf positions and their changes is indispensable to the accurate detection of coastal polynyas.

In this study, we propose a new SSM/I algorithm that distinguishes between fast ice and thin ice and estimates thin ice thickness for the purpose of investigating Antarctic coastal polynyas. The algorithm is applicable to the whole Antarctic Ocean, and provides twice-daily estimates of thin ice thickness with a spatial resolution of 12.5 km. We develop the algorithm from which the thickness can be used toward the estimation of heat flux and sea ice production. Section 2 describes the data used in this study. Section 3 describes the method that estimates sea ice thickness from the AVHRR data. Section 4 presents the algorithm that detects thin ice regions and estimates thin ice thicknesses from the SSM/I data, with a discussion of atmospheric effects. Section 5 describes the method of the detection of fast ice and ice shelves. Section 6 discusses the validity of the algorithm and presents a distribution map of thin ice occurrences. Section 7 provides our remarks.

2. Data

In this study, we use the twice-daily (ascending and descending orbits) Equal Area Scalable Earth-Grid (EASE-Grid) SSM/I brightness temperature data (1992–2001) provided by the National Snow and Ice Data Center (NSIDC) in Boulder, Colorado. The 85- and 37-GHz (both vertical and horizontal polarization) channel data are used to take advantage of both the higher spatial resolution at 85 GHz (~12.5 km; resolution at 37 GHz is ~25 km) and the lower sensitivity to water vapor at 37 GHz. Figure 2 shows their grids schematically. The 37-GHz data are interpolated onto the 85-GHz grid as follows: when an 85-GHz grid cell overlaps with two or four 37-GHz grid cells (e.g., black circles in Fig. 2), we average these. When an 85-GHz grid cell lies within a 37-GHz grid cell (e.g., the black triangle in Fig. 2), we use the 37-GHz value for the 85-GHz grid cell. Missing data pixels on the SSM/I map are filled using spatial (and if needed temporal) interpolation when we show hemispheric maps and time series (as will be shown later).

National Oceanic and Atmospheric Administration (NOAA) AVHRR channel-4 and -5 data (1992–2001) are provided by the National Institute of Polar Research of Japan and NSIDC. The spatial resolution is ~1.25 km. These data were received at the Syowa, McMurdo, Palmer, and Casey Stations (Fig. 1). Station coverage ranged from several hundred to several thousand kilometers.

The 40-yr European Centre for Medium-Range Weather Forecasts (ECMWF) Re-Analysis (ERA-40) data for 1992–2001 are used to provide air temperatures at 2 m, dewpoint temperatures at 2 m, winds at 10 m, and surface mean sea level pressures. The spatial resolution is 1.125° × 1.125°. The atmospheric data are used

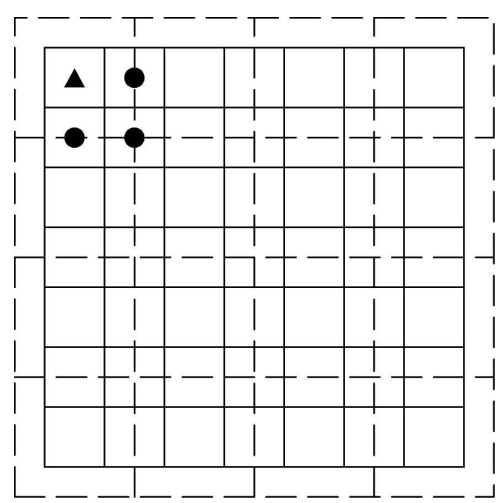

FIG. 2. A schematic illustration of the SSM/I EASE-Grids. The solid and dashed lines indicate the 85-GHz (12.5 km) and 37-GHz (25 km) grids, respectively. Black circles indicate the 85-GHz grid cells that overlap with two or four 37-GHz grid cells. A black triangle indicates an 85-GHz grid cell that is included within a 37-GHz grid cell.

for the heat flux calculation to estimate ice thickness using AVHRR data in the next section. Since the data are available every 6 h, they are averaged over 24 h to minimize the effects of thermal inertia described in the next section.

3. Estimation of the thermal ice thickness from AVHRR data

Yu and Rothrock (1996) and Drucker et al. (2003) provided a method of ice thickness estimation by comparing ice thicknesses obtained from the upward-looking sonar (ULS) data with AVHRR data in a seasonal sea ice zone of the Northern Hemisphere. Their estimation is applicable to sea ice with thicknesses <0.5 m (Yu and Rothrock 1996). Tamura et al. (2006) confirmed the validity of the method by using ice thickness data obtained from in situ observations in an Antarctic coastal polynya. The present study follows their method. The method estimates ice thickness from conductive heat through sea ice calculated from the surface temperature retrieval, assuming thermal equilibrium and a constant ocean temperature taken to be at the freezing point (271 K). This ice thickness is hypothetical, because the total heat loss is calculated under the assumption of a uniform ice thickness for a given grid cell, whereas in reality the ice thickness is not uniform. Because this thickness is not a simple arithmetical average, following Drucker et al. (2003), this study calls this thickness the *thermal ice thickness.*

We estimate the AVHRR thermal ice thickness from the conductive heat flux calculated from the residual of the sum of radiative and turbulent fluxes, where the conductivity of sea ice is assumed to be 2.03 W m^{-1} K^{-1}. For simplicity, we neglect snow cover, because new ice generally does not have a significant amount of snow. Radiative and turbulent heat fluxes between the air and the sea ice surface are calculated with empirical and bulk formulas similar to the calculations of Nihashi and Ohshima (2001). Ice surface temperatures are obtained using AVHRR channel-4 and -5 data with the empirical equation found in Key et al. (1997). We choose only night cases to avoid the ambiguity created by shortwave radiation absorption into the ice interior (Grenfell and Maykut 1977). We also choose only cases that are free from cloud cover, ice fog, and water vapor through a visual inspection of AVHRR channel-4 imagery.

Our calculation assumes that the heat exchange between the sea ice and atmosphere occurs immediately with the thermal inertia of ice completely neglected. The percentage contribution of the thermal inertia component to the net heat flux is only ~8% when the air temperature is assumed to change 10 K in 24 h (a typical change) for thin ice with a thickness of 0.1 m under typical meteorological conditions along the winter Antarctic coast (air temperature is 258 K and net heat flux is 250 W m^{-2}). Thus, the thermal inertia is neglected in this study.

For the purpose of comparing the AVHRR data with the SSM/I data, the AVHRR 1.25-km gridded thermal ice thickness data are mapped onto the 12.5-km SSM/I grid. One SSM/I grid cell contains several tens of AVHRR grid cells with various thermal ice thicknesses. Because the variability of surface fluxes is nonlinear with respect to that of ice thickness, the arithmetic average of the AVHRR thickness is slightly different from the hypothetical thermal ice thickness suitable for the heat loss calculation. For the SSM/I grid cell, we also use the hypothetical thermal ice thickness for which the total heat loss can be calculated under the assumption of uniform ice thickness. This AVHRR thermal ice thickness will be used as comparison and validation data for the estimation of thin ice thickness from the SSM/I data in the next section. Figures 3a,b are examples of the spatial distribution of the AVHRR surface temperature drawn from the original AVHRR grid. Figures 3c,d are those of the thermal ice thickness mapped onto the SSM/I grid in the regions around the Ross Ice Shelf and Mertz Glacier Tongue.

4. Thin ice thickness algorithm

a. Estimation of thin ice thickness

In this subsection, we present an algorithm that detects thin ice and estimates the ice thickness from SSM/I data. The basis of the algorithm stems from the microwave characteristics of thin ice as determined from both experiments and observations. From past studies (e.g., Cavalieri 1994; Martin et al. 2004), PR values at a given frequency decrease as sea ice thickness increases from new to young to first-year ice. Although the skin depth of bare sea ice at 85 GHz is about several millimeters (Ulaby et al. 1982), microwave brightness temperatures correlate with the surface salinity (brine volume fraction; Vant et al. 1978), which is sensitive to thin ice thickness (Cox and Weeks 1974; Kovacs 1996). Thus, we think that we can detect bare ice with thicknesses less than 0.1–0.2 m from PR similar to the approach taken by Martin et al. (2004). Because of this dependency, PR values at 85 and 37 GHz (PR_{85} and PR_{37}) are used as the main parameters in the algorithm. The PR_{85} (Figs. 4a,b) and PR_{37} (Figs. 4c,d) values are high in the polynya areas (Ross Sea coastal polynya and Terra Nova Bay Polynya in Figs. 4a,c and Mertz Glacier

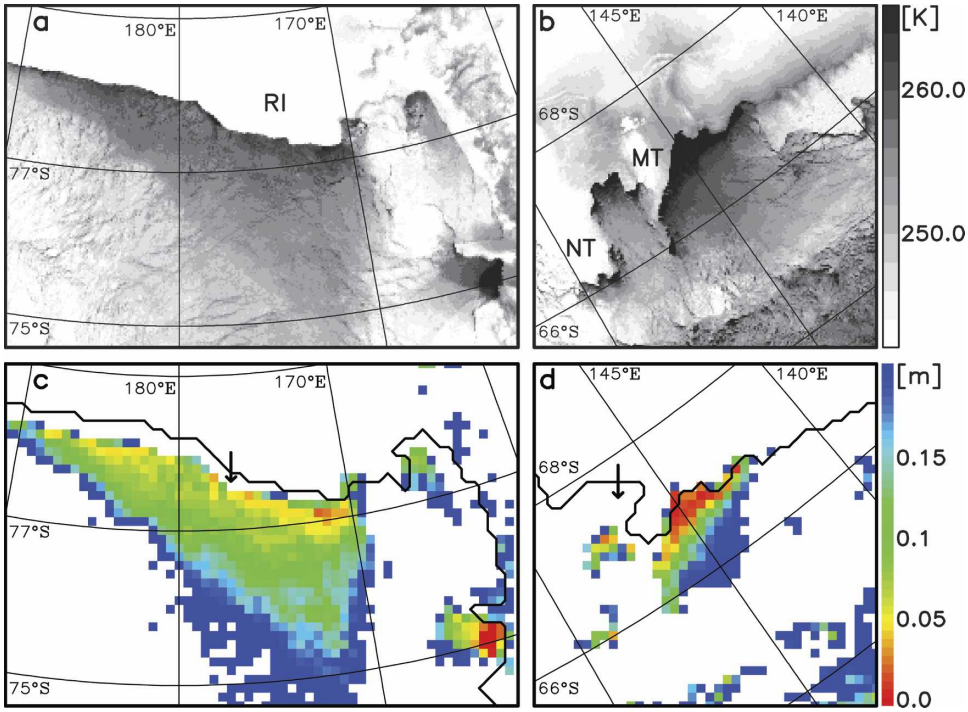

FIG. 3. (a), (b) AVHRR ice surface temperature imagery and (c), (d) AVHRR thermal ice thickness for (a), (c) the region around the Ross Ice Shelf on 10 Aug 1992 and for (b), (d) the region around the Mertz Glacier Tongue on 19 Aug 2000. RI is the Ross Ice Shelf, MT is the Mertz Glacier Tongue, and NT is the Ninnis Glacier Tongue. (c), (d) The solid lines indicate the coastline of the SSM/I landmask. (c), (b) The white pixels show the area of thick ice with thicknesses of >0.5 m, ice shelf, and continental ice. Black arrows designate those parts of the ice shelf and glacier tongues that are discussed in section 5.

Polynya in Figs. 4b,d) relative to the surrounding pack ice.

To derive a relationship between the AVHRR thermal ice thicknesses and the SSM/I PR values, we examine 13 polynya events (two from the Weddell Sea coastal polynyas, two from the Ross Sea coastal polynya, five from the Mertz Glacier Polynya, and four from the Cape Darnley Polynya). These 13 cases are carefully selected among several hundreds of images for their clearness and uniformity of the coastal polynya. These 13 cases span a period of time ranging from 3 April to 2 November with wind speed, air temperature, and humidity in the range of 4–14 m s^{-1}, 242–265 K, and 69%–99%, respectively. Because the time period of this study covers the freezing period, the algorithm results are valid for the freezing period only. Further, we only use those SSM/I grid cells in which the AVHRR thermal ice thicknesses are nearly uniform. In these cells, more than 80% of the AVHRR thermal ice thicknesses are within ±0.05 m of the averaged value of that cell.

Figure 5a shows a scatterplot of the AVHRR thermal ice thicknesses versus the corresponding PR_{85} for all 13 cases. The standard deviation of the AVHRR thermal ice thicknesses within the SSM/I grid cells of PR_{85} is ~0.020 m on average for all the plotted data. A principal component analysis was made of the plotted data and the principal component axis is indicated by the heavy black line in Fig. 5a. With respect to the principal component axis of this plot, the explained variance of the principal component to the total sample of 1315 is 96%, and the standard deviation of the difference between the AVHRR thermal ice thickness and the principal component axis is <0.04 m. On the other hand, for all the data whose AVHRR thermal ice thickness is >0.2 m (data beyond the figure), the mean PR_{85} value is calculated to be ~0.027 with the standard deviation of ~0.0075. For all the data whose AVHRR thermal ice thickness is >0.2 m, the probability that PR_{85} is >0.0495, which is the mean value plus 3 times the standard deviation, is ~2%. This PR value of 0.0495 (the dotted line in Fig. 5a) corresponds to the AVHRR thermal ice thickness of 0.1 m from the principal component axis. Thus, the probability that thick ice with thickness >0.2 m is misinterpreted as thin ice with thickness <0.1 m is 2%. In this study, we use this principal com-

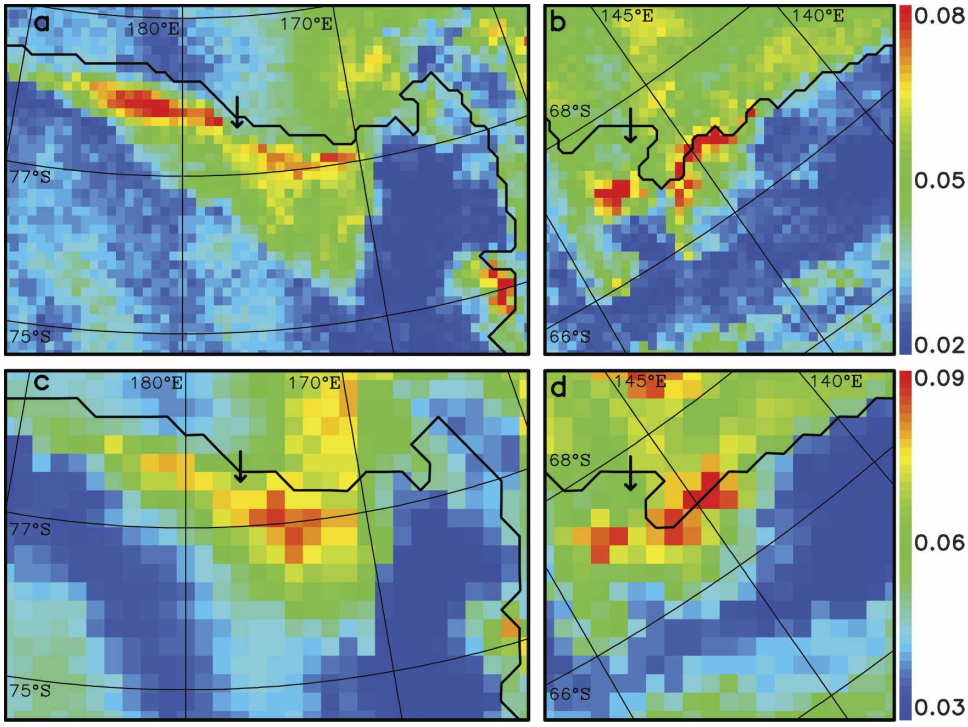

FIG. 4. (a), (b) Spatial distribution of polarization ratio of SSM/I 85 GHz and (c), (d) polarization ratio of SSM/I 37 GHz for (a), (c) the region around the Ross Ice Shelf on 10 Aug 1992 and for (b), (d) the region around the Mertz Glacier Tongue on 19 Aug 2000. Indications of the lines and arrows are the same as those in Fig. 3.

ponent axis to estimate thin ice thickness from PR_{85} for ice thicknesses of <0.1 m.

A similar analysis is also applied to the PR_{37} data. Figure 5b is the scatterplot of AVHRR thermal ice thicknesses and the corresponding PR_{37} for all 13 cases. The standard deviation of the AVHRR thermal ice thicknesses within the SSM/I grid cells of PR_{37} is ~0.029 m on average for all the plotted data. For PR_{37}, the explained variance of the principal component to the total sample of 277 is 99%, and the standard deviation of the difference between the AVHRR thermal ice thickness and the principal component axis is ~0.07 m. On the other hand, for all the data whose AVHRR thermal ice thickness is >0.4 m (data beyond the figure), the mean PR_{37} value is ~0.0346 with the standard deviation of ~0.0075. For all the data whose AVHRR thermal ice thickness is >0.4 m, the probability that PR_{37} is >0.0571, which is the mean value plus 3 times the standard deviation, is ~2%. This PR value of 0.0571 (the dotted line in Fig. 5b) corresponds to the AVHRR thermal ice thickness of 0.2 m from the principal component axis. Thus, the probability that thick ice with thickness >0.4 m is misinterpreted as thin ice with thickness <0.2 m is 2%. We use this principal component axis to estimate thin ice thickness from PR_{37} for ice thicknesses of 0.1–0.2 m.

The relationship between SSM/I PR values and physical thin ice thicknesses may not actually be linear (Wensnahan et al. 1993). However, the thermal ice thickness that we seek may not necessarily show the same relationship as that of physical ice thickness. In this study, we make use of the result obtained from the comparison of SSM/I PR values with ice thicknesses derived from AVHRR data, which shows a linear-like relationship, although with relatively large scatter (Fig. 5). The dashed curve in Fig. 5b indicates the relationship derived from the Martin et al. (2004) algorithm for an Arctic polynya. This curve apparently does not fit our plot and the extent of thin ice regions would be overestimated if this curve were used for ice thickness estimations. The relationship between SSM/I PR values and thin ice thicknesses may have a hemispheric or latitudinal bias due to differences in atmospheric conditions.

Tamura et al. (2006) examined the relationship between the PR values and the AVHRR thermal ice thicknesses in an Antarctic coastal polynya, where ice thickness data from in situ measurements were simul-

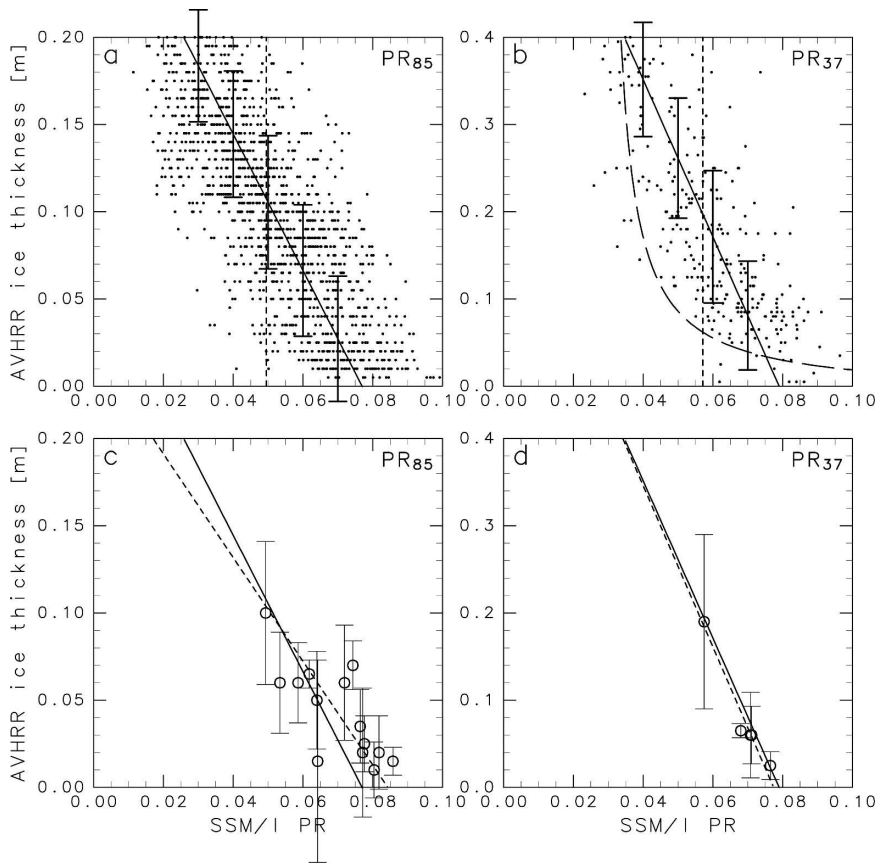

FIG. 5. Scatterplot of AVHRR thermal ice thickness on the vertical axis and SSM/I PR for (a), (c) 85 GHz and (b), (d) 37 GHz on the horizontal axis. The solid line denotes the principal component axis of these plots. (a), (b) The vertical lines with crossbars show the standard deviation of the plots with respect to the principal component axis line. (c), (d) The open circles denote the result of Tamura et al. (2006). (c), (d) The vertical lines with crossbars show the standard deviation of AVHRR thermal ice thicknesses in the SSM/I grid cell for each plot data.

taneously obtained. In Figs. 5c,d, the scatterplot (open circles) and the principal component axis from their study (dashed lines) are superimposed on the principal component axis of Figs. 5a,b. The vertical lines with crossbars show the standard deviation of AVHRR thermal ice thicknesses in the SSM/I grid cell of each plot. Although the data are limited and the standard deviations are not small, the results of the present study are consistent with those of Tamura et al. (2006).

Based on the relationships shown in Figs. 5a,b, we propose the following algorithm for thin ice detection. First, pixels with PR_{85} values >0.0495 are regarded as thin ice pixels with thicknesses in the range of 0–0.1 m, and the thickness h is calculated using Eq. (1). Second, for the remaining pixels, pixels with PR_{37} values >0.0571 are regarded as thin ice pixels with thicknesses in the range of 0.1–0.2 m, and the thickness h is calculated using Eq. (2):

$$h = -3.912(PR_{85}) + 0.3010 (PR_{85} \geq 0.0495 \text{ for } 0.0\text{–}0.1 \text{ m}). \quad (1)$$

$$h = -9.020(PR_{37}) + 0.7125 (PR_{37} \geq 0.0571 \text{ for } 0.1\text{–}0.2 \text{ m}). \quad (2)$$

For the case when the thickness is calculated to be <0 m in Eq. (1) (PR_{85} value is >0.077), the thickness is assumed to be 0.01 m. For the case when the thickness is calculated to be <0.1 m in Eq. (2) (PR_{37} value is >0.068), the thickness is assumed to be 0.1 m. This algorithm has discontinuities at the threshold of 0.1 m, which is a limitation of the algorithm. Practically, thinner ice areas in coastal polynyas near the coast are usually only detected with higher-resolution 85-GHz data. Relatively thicker ice areas of the polynya farther from the coast are often detected with the low-resolution 37-GHz data. There is a difference between ice thicknesses estimated from PR_{85} and PR_{37} for the

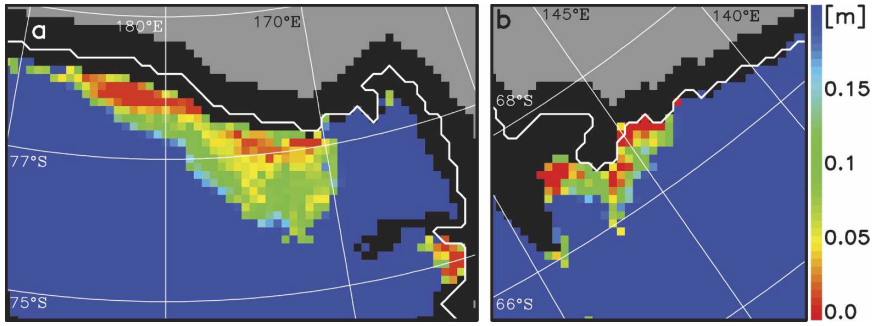

FIG. 6. Spatial distribution of the ice thickness estimated from the algorithm of SSM/I, with black masked pixels showing fast ice regions for (a) the region around the Ross Ice Shelf on 10 Aug 1992 and for (b) the region around the Mertz Glacier Tongue on 19 Aug 2000. The white solid lines indicate the coastline of the SSM/I landmask. The gray pixels show the area of ice shelf and continental ice.

same pixel as a result of footprint differences and the effect of interpolation. In addition to these differences, ice thicknesses derived from PR_{85} may be overestimated due to atmospheric effects, which will be discussed in the next subsection.

When the thin ice algorithm is applied to the whole Antarctic Ocean, we distinguish sea ice pixels from open water pixels by using the 15% ice concentration threshold from the NT2 algorithm (Markus and Cavalieri 2000). Similar results are obtained when the Bootstrap algorithm (Comiso et al. 1984; Comiso 1986) is used. The thin ice algorithm is then applied only to the sea ice pixels. Sea ice pixels, with thicknesses estimated by the algorithm to be >0.2 m, are labeled as first-year ice. Figures 6a,b are examples of the spatial distribution of ice thickness estimated from the SSM/I thin ice algorithm in the regions around the Ross Ice Shelf and Mertz Glacier Tongue, respectively. The Ross Sea coastal polynya, Terra Nova Bay Polynya, and Mertz Glacier Polynya are detected as thin ice regions with thicknesses <0.2 m. The shape and location of thin ice (polynya) regions generally coincide with those of AVHRR thermal ice thickness (Figs. 3c,d). Fast ice designated by black in Figs. 6a,b will be discussed in section 5.

b. *Atmospheric effects*

Microwave data are not affected very much by the atmosphere along the Antarctic coastal regions during winter because the atmosphere is relatively dry at high latitudes. However, when a midlatitude weather system with a comparatively high water vapor content is advected southward (Fig. 7a), the 85-GHz data are affected by the disturbance to some extent (Fig. 7b). On the other hand, the 37-GHz data are much less affected by the atmospheric effects (Fig. 7c). Because the PR_{85} is the main parameter in the algorithm, either a correction for atmospheric effects or the exclusion of those pixels impacted by these effects is needed.

Figure 8a shows a scatterplot of PR_{85} versus PR_{37} for those 36 cases with no cloud cover over sea ice in the region around the Cape Darnley Polynya, after visual screening of clouds from AVHRR channel-4 images. Figure 8b is a similar plot but for those 21 cases with atmospheric disturbances over sea ice. In both cases, we use only the data where the 85-GHz 12.5-km grid cell is completely included inside the 37-GHz 25-km grid cell (see section 2). On clear days, PR_{85} and PR_{37} have a proportional relationship although there is considerable scatter due to the footprint differences between the two. On cloudy days, the ratio of PR_{85} to PR_{37} is smaller overall than on clear days. This implies that ice thickness tends to be overestimated from PR_{85} on cloudy days. A semiempirical radiative-transfer model (Gloersen and Cavalieri 1986) is one way of correcting for atmospheric effects. In this method, the bias of PR_{85} calculated from the model will be corrected relative to PR_{37}, which is less sensitive to water vapor. If we evaluate the atmospheric effects on PR_{85} from the relationship between thin ice thicknesses estimated from PR_{37} and PR_{85}, the bias of PR_{85} could be corrected to some extent. However, even on clear days, the relationship between PR_{85} and PR_{37} has an error due to their footprint differences (on the plot of clear days, the standard deviation of the difference between PR_{85} and the least squares line of PR_{85} and PR_{37} is $\pm 9.5 \times 10^{-3}$). Further, the thickness estimation from PR_{37} also has an error, which is represented by the standard deviation of the difference from the principal component axis as shown in Fig. 5b [the value converted from this error to PR_{85} through Eq. (1) is $\pm 1.8 \times 10^{-2}$]. These errors are comparable to or larger than the bias of PR_{85} due to the

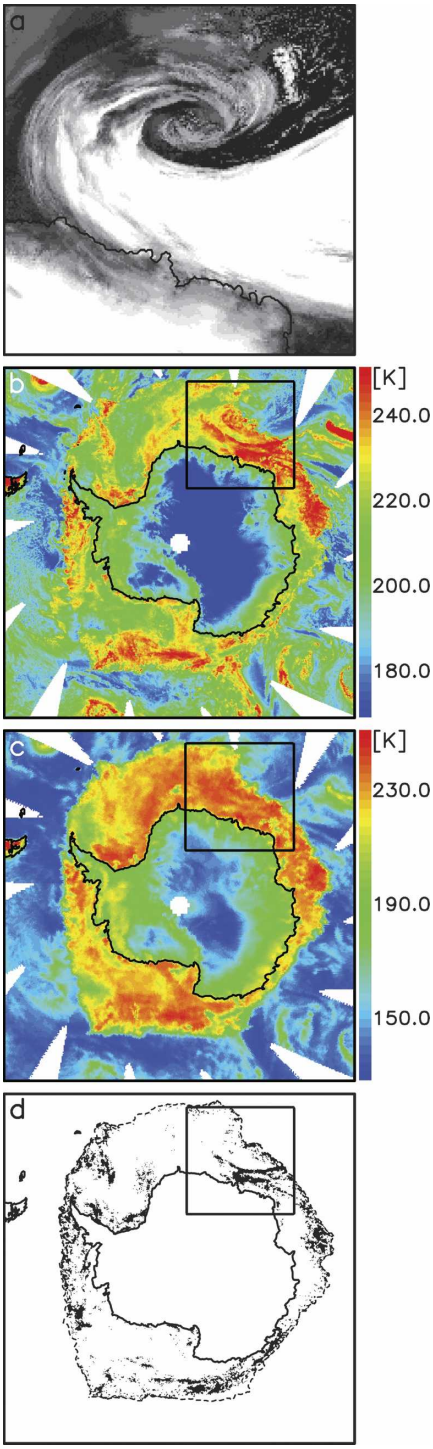

FIG. 7. Comparison of the AVHRR and SSM/I imagery on 1 Sep 2000. (a) The AVHRR channel-4 image. (b) The horizontal polarization brightness temperature of SSM/I 85 GHz with white masked pixels showing the missing data of SSM/I. (c) The horizontal polarization brightness temperature of SSM/I 37 GHz with white masked pixels of missing data. (d) Pixels (masked by black) regarded as areas affected by water vapor with the dashed line outlining the boundary of the 15% sea ice concentration. The solid lines indicate the coastline of the SSM/I landmask. The rectangular box denotes the region in (a).

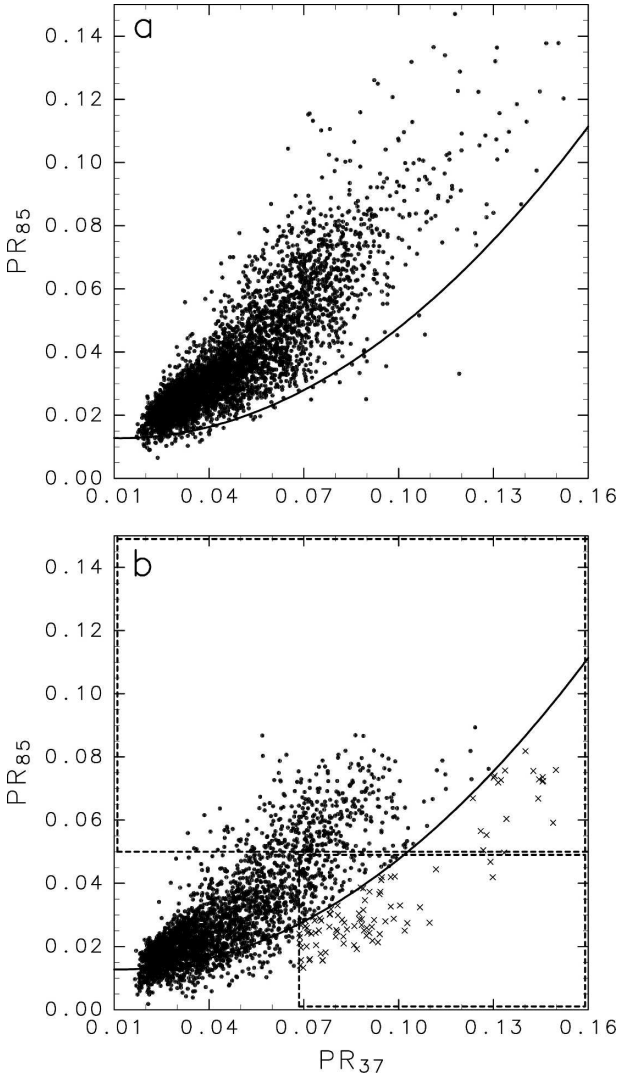

FIG. 8. Scatterplot of PR_{85} vs PR_{37} for (a) clear and (b) cloudy days. The solid curve indicates the boundary of the plots regarded as an area affected by water vapor. (b) The two rectangular boxes indicate the areas of the rejected pixel due to water vapor. Pixels that are below the solid curve and within the two rectangular boxes are rejected. Actual rejected data are designated by the × symbols in (b).

atmospheric effects (for a thin ice area of thickness <0.2 m, the difference between the least squares line of PR_{85} and PR_{37} on clear days and that on cloudy days is 1.0–1.5 × 10^{-2}). In summary, we could not find a satisfactory method of making a reliable correction for atmospheric effects, at least from this analysis.

Instead, we propose to eliminate those pixels that suffer from atmospheric effects by the following method. The method uses the relationship between PR_{85} and PR_{37}. In the scatterplot on clear days (Fig. 8a), data for the PR_{37} value of 0.02–0.15 are segmented

into fifteen 0.005 intervals, and the mean value and standard deviation of PR_{85} are calculated for each segment. Then a least squares–best-fit curve is drawn with the least squares fitting to the mean values minus twice the standard deviation for each segment (solid curve in Fig. 8). This curve is represented by the equation

$$PR_{85} = 4.492(PR_{37})^2 - 0.1062(PR_{37}) + 0.013\,36. \tag{3}$$

Only ~2% of all the data on clear days are below this curve. On the other hand, when this curve is applied to cloudy days, ~18% of all the data are below this curve. In this study, the data below this curve are regarded as those pixels being affected by water vapor. Figure 7d shows an example of the spatial distribution of those pixels that fall below the curve represented by Eq. (3). The percentage of these pixels is ~16% for the whole Antarctic sea ice region on an annual average. But we do not have to reject all these pixels. We should consider atmospheric effects for only the thin ice pixels as estimated from PR_{85} within the following two ranges. One is the upper rectangular box in Fig. 8b: from PR_{85}, ice thickness is determined to be <0.1 m but may be overestimated. The other is the lower rectangular box in Fig. 8b: from PR_{85}, ice thickness is determined to be >0.1 m, but actually should be <0.1 m as determined from PR_{37} under the assumption that PR_{37} is not affected by water vapor. Thus, pixels below the curve defined by Eq. (3) in the two rectangular boxes are rejected in this algorithm. Most of the rejected data (the × symbols in Fig. 8b) exist in marginal ice zones at relatively low latitudes with higher atmospheric water vapor. The percentage of actual rejected data is ~7% for the whole Antarctic sea ice region on an annual average. Because the 85-GHz data are much less affected near the Antarctic continent, the 85-GHz algorithm does not have a serious problem with atmospheric effects for Antarctic coastal polynyas. To make a seamless dataset of ice thickness, spatial (and if needed temporal) interpolation is applied. For those pixels with ice thicknesses of >0.1 m after the interpolation, the thicknesses should be reduced to 0.1 m because it has already been determined from PR_{37} that the thicknesses are <0.1 m.

5. Detection of fast ice

In addition to first-year ice and thin ice, there are other types of ice along the Antarctic coastline including ice originating from sea ice such as landfast ice and ice originating from continental ice such as ice shelves, glacier tongues, and icebergs. According to the World Meteorological Organization (1970), landfast ice, glacier tongues, grounded icebergs, and old ice accumulated around them are defined as fast ice. In this study, we consider these ice types and also ice shelves close to the coastline as fast ice for convenience hereafter. It is difficult to distinguish fast ice from thin ice because both have similar PR signatures. Retrievals from both the NASA Team and Bootstrap algorithms, for example, misclassify part of the Ross Ice Shelf as low concentration sea ice (Comiso and Steffen 2001). Coastal polynyas, such as the Mertz Glacier Polynya, are often formed near or next to fast ice, particularly around the East Antarctic coast.

The extent of landfast ice may change by its breakup and regeneration. The shape and location of ice shelves and icebergs also change due to calving and drifting. Their positions in the SSM/I 12.5-km grid may change by up to several pixels per month. Monitoring the position of fast ice is indispensable for the accurate detection of coastal polynyas. A comparison of Fig. 3c with Figs. 4a,c shows that the area (indicated by arrows in each figure) of high PR values, which are comparable to that of thin ice, is the edge of the Ross Ice Shelf. A comparison of Fig. 3d with Figs. 4b,d shows that the area (also indicated by arrows in each figure) of high PR values is an area of fast ice that includes the Mertz and Ninnis Glacier Tongues. Landfast ice also has PR values similar to that of thin ice.

Glacier tongues and icebergs, which originate from land ice, have similar characteristics to that of continental ice. Figure 9 is an example of the 85-GHz vertical versus horizontal polarization brightness temperature scatterplot for thin sea ice, fast ice, and continental ice. This figure illustrates that although the PR values of fast ice and continental ice are similar to those of thin sea ice, the fast ice and continental ice have lower brightness temperatures than thin sea ice. It should also be noted that the fast ice and continental ice clusters coincide.

The following algorithm is proposed for the detection of fast ice, based on the above-mentioned microwave characteristics. On the scatterplot of the 85-GHz vertical and horizontal polarization brightness temperatures for only continental ice pixels within ~250 km from the coast, we calculate the first and second principal component axes and the standard deviation for each component. Then an ellipse is drawn from the center of gravity of the cluster of points, where the first and second principal component axes are set to the major and minor axes, respectively, with the amplitude being 2.5 times the standard deviation. About 96% of the continental ice pixels exist inside this ellipse (see Fig. 9). Fast ice detection is conducted on a monthly basis using this

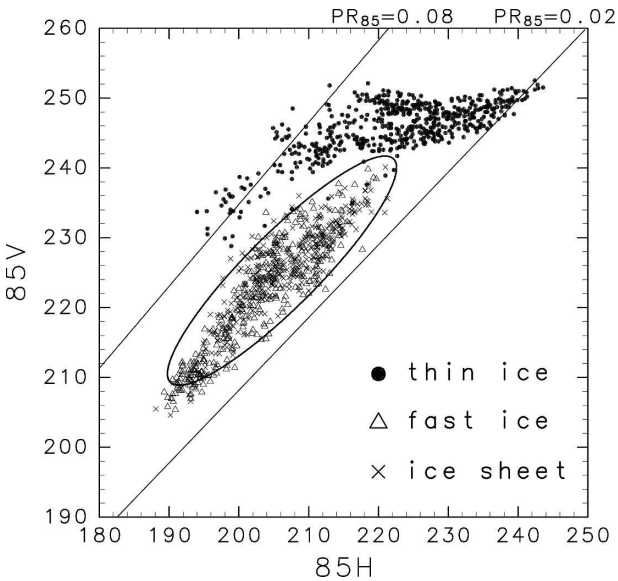

FIG. 9. Scatterplot of vertical vs horizontal polarization brightness temperatures of 85 GHz for thin ice, fast ice, and continental ice near the Lützow-Holm Bay in July 2000. The two solid lines show PR_{85} values of 0.02 and 0.08. In this plot, 95% of continental ice data exists inside the ellipse.

ellipse. We calculate the ellipse twice a day (ascending and descending orbits) for three consecutive months. Then for each pixel, we count the frequency with which the brightness temperatures are plotted inside the ellipse of each scene for the three months (~180 scenes). Pixels that have a frequency of more than 70% are regarded as fast ice pixels. Because the probability that the fast ice pixels are not inside the ellipse is 4% for each scene, the detection of fast ice using only a small number of data is difficult. For the reliable detection of fast ice more data are needed. Thus we make a fast ice map on a monthly basis using the data for three consecutive months including the previous and following months. A time scale of one month is sufficient to detect the change of fast ice at the SSM/I spatial resolution. This method is also applied to land areas within ~50 km from the coast to obtain updated coastlines given the dynamic nature of ice shelves and ice tongues. Black masked pixels in Figs. 6a,b are fast ice areas detected by this algorithm. The Ross Ice Shelf and the fast ice including the Mertz and Ninnis Glacier Tongues are clearly detected. These areas would be classified as thin ice regions based only on their PR values (see Figs. 4a–d).

For validation, we compare the results with AVHRR channel-4 images in the sea area where fast ice detection is particularly difficult. Lützow-Holm Bay is often covered with landfast ice, and the breakup and regeneration of landfast ice has repeatedly occurred since the 1990s. In addition, because of the relatively low latitude, the brightness temperatures of continental ice and fast ice tend to have properties similar to those of thin ice due to surface layer warming in summer. Therefore, fast ice detection in this region is relatively difficult compared to other areas. To examine the seasonal change, we selected the following AVHRR images: June 1997 when the breakup just started (Fig. 10a), February 1998 when the breakup ended (Fig. 10b), and June 1998 when the coastal region was covered with first-year ice (Fig. 10c). These cases are compared with the results obtained from the fast ice detection algorithm. The results of the algorithm for the landfast ice (Fig. 10d), the breakup of landfast ice (Fig. 10e), and the accumulation of first-year ice (Fig. 10f) are in generally good agreement with the corresponding AVHRR images. Only the landfast ice that had existed continuously at least for several months is detected as fast ice.

Monthly fast ice detection cannot catch drifting icebergs appropriately due to their relatively high-speed motion. The detection on shorter time scales is necessary. In this study, we show an example of iceberg detection for the case of large icebergs that were created by the large calving event that occurred at the Ross Ice Shelf in 2000. Martin et al. (2007) examined the drift of these same icebergs in the Ross Sea from Moderate Resolution Imaging Spectroradiometer (MODIS) and RADARSAT ScanSAR imageries. To detect drifting icebergs, we make an ellipse similar to the one in Fig. 9 for a more confined region (the rectangular box of Ross Sea coastal polynya in Fig. 1). The detection is conducted on a 5-day basis, and pixels that have a frequency of more than 70% for the plot being inside the ellipse are regarded as fast ice (iceberg) pixels. Figure 11 is an example of the result obtained from this iceberg detection method. From the comparison with the corresponding AVHRR images, large icebergs that can be resolved by the SSM/I grid scale are found to be detected to some degree. In conclusion, the fast ice detection algorithm presented in this study can detect to some extent the position of ice shelves, landfast ice, glacier tongues, and large icebergs, and their changes in location within the SSM/I grid scale.

6. Validation and application

Table 1 compares the AVHRR thermal ice thickness with the SSM/I ice thickness derived from the thin ice algorithm for each region and season. All the cases in Table 1 except the case of the marginal ice zone (MIZ) correspond to the 13 polynya events that are used in plotting Fig. 5. The biases between the two (AVHRR

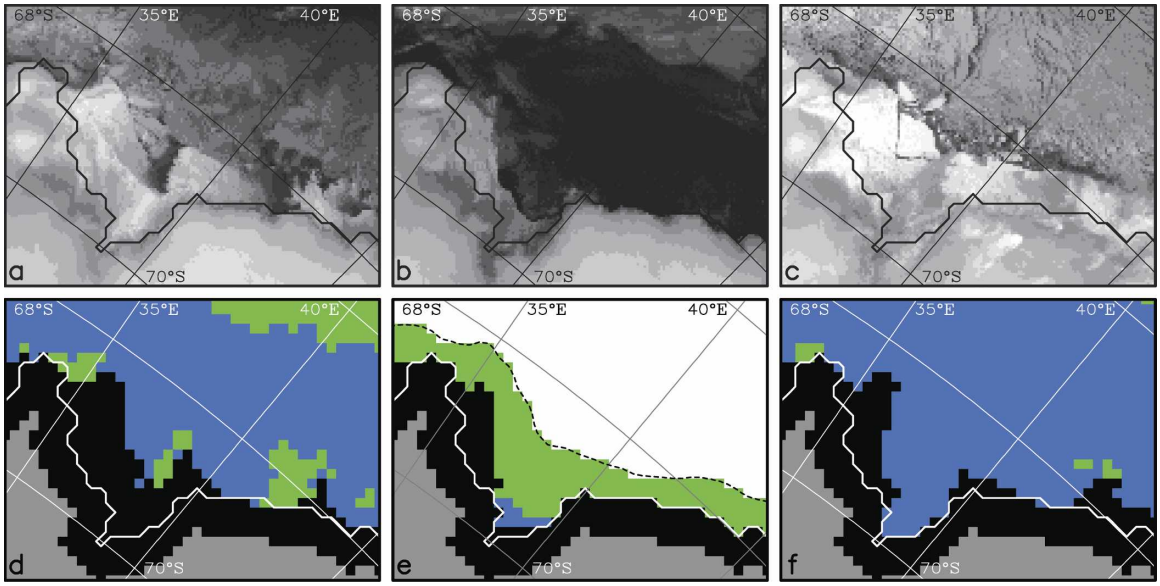

FIG. 10. Comparison of (a)–(c) the AVHRR channel-4 imagery and (d)–(f) ice type map derived from the SSM/I algorithm for the Lützow-Holm Bay on (a), (d) 22 Jun 1997; (b), (e) 13 Feb 1998; and (c), (f) 22 Jun 1998. The black pixels show fast ice, the blue ones show first-year ice, the green ones show thin ices of <0.2 m, and the gray ones show ice shelf and continental ice. The dashed line outlines the boundary of the 15% sea ice concentration. The solid lines indicate the coastline of the SSM/I landmask.

minus SSM/I) and the standard deviations of the difference between the two are listed. This table suggests that the algorithm gives a small bias, independent of region and season, if the AVHRR thermal ice thickness is assumed to be true. The accuracy (standard deviation) of the SSM/I ice thickness retrievals is ~0.05 m on average, independent of region and season. From the results shown in Table 1, this algorithm should be applicable to the entire Antarctic Ocean.

An exact error estimation of the algorithm is only possible from the comparison with in situ measurement data, which are not available at present. Instead, we

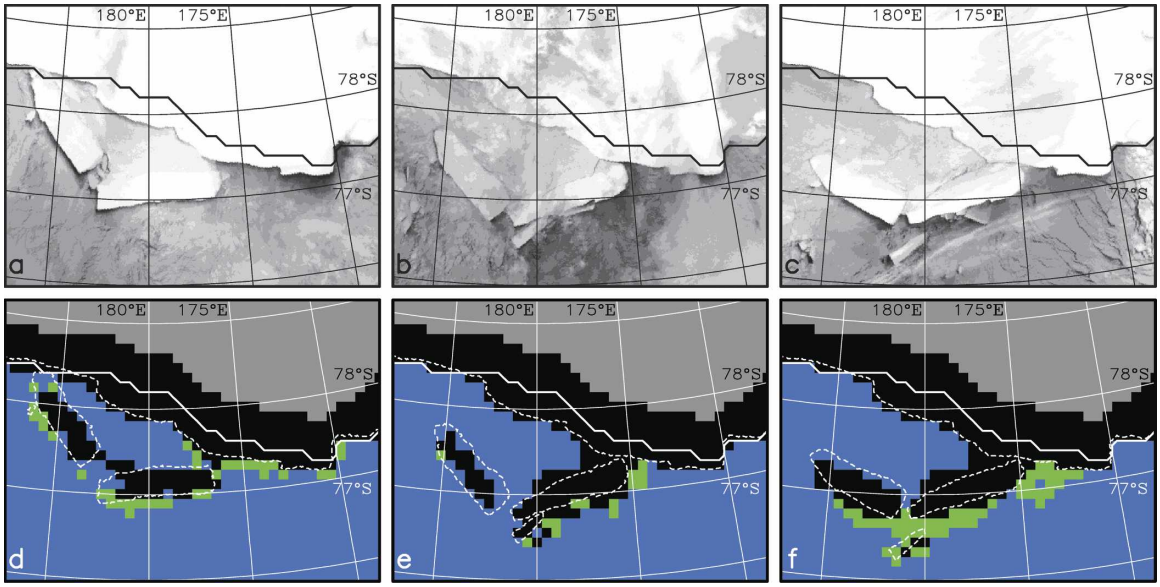

FIG. 11. Comparison of (a)–(c) the AVHRR channel-4 imagery and (d)–(f) ice type map derived from the SSM/I algorithm for the region around the Ross Sea coast on (a), (d) 27 Jul 2000; (b), (e) 10 Aug 2000; and (c), (f) 17 Aug 2000. Indication of colored pixels and lines are the same as those in Fig. 10. The dashed line shows the edge of the ice shelf and icebergs inferred from the AVHRR data.

TABLE 1. The bias of the AVHRR minus SSM/I thin ice thicknesses and the standard deviation (STD) of the difference between the two with the sample number. These are calculated for each of five regions and three seasons independently.

Place/season	Cape Darnley	Ross	Weddell	Mertz	March–May	June–August	September–November	MIZ(Adélie)
Bias	−0.04 cm	+0.06 cm	+1.16 cm	+0.34 cm	−0.06 cm	+0.48 cm	+0.03 cm	−1.60 cm
STD	4.7 cm	5.0 cm	6.3 cm	4.7 cm	4.8 cm	4.8 cm	5.0 cm	5.1 cm
Sample No.	610	89	64	483	406	456	339	187

estimate the error of the algorithm from the comparison with the AVHRR thermal ice thicknesses from the 13 polynya cases listed in Table 1. The probability that AVHRR thermal ice thicknesses of >0.2 m are misclassified as thin ice of thickness <0.2 m from the algorithm is 3.0%. The probability that AVHRR thermal ice thicknesses of <0.2 m are misclassified as fast ice or first-year ice is 6.8%. The algorithm distinguishes thin ice from thick ice and from fast ice with an accuracy of ~95% based on the 13 polynya events, although the algorithm still has some errors of several centimeters for specific ice thickness estimates. The probability that AVHRR thermal ice thicknesses in the range of 0.1–0.2 m are misclassified as ice of thickness <0.1 m from PR_{85} is 17%. The probability that AVHRR thermal ice thicknesses of <0.1 m are misclassified as ice of thickness in the range 0.1–0.2 m from PR_{37} is 16%.

Figure 12 shows a time series of the offshore wind speed and the polynya extent of the Ross Sea coastal polynya. Daily wind data are derived from ERA-40 data at the location of 76.5°S and 174.4°E (designated by the × symbol in Fig. 1). This figure demonstrates that the opening and closing of the coastal polynya corresponds well to the increase and decrease of the offshore wind speed. When we calculate the lag correlation between the polynya extent and the offshore wind component, the highest correlation occurs at one day lag of the polynya extent. Its correlation coefficient is 0.65. In the same way, we also compare the polynya extent with the offshore wind component in the Weddell Sea coastal polynyas, the Mertz Glacier Polynya, and the Cape Darnley Polynya. In all cases, the two correspond well to each other and the correlation coefficients are greater than 0.5. These results further support the validity of the detection of thin ice regions with this algorithm, because larger coastal polynyas are formed by stronger offshore winds.

We next compare the algorithm of the present study with the PSSM (Markus and Burns 1995), a different approach using passive microwave data. The PSSM algorithm is based on simulating microwave images of polynya events by convolving an assumed brightness temperature distribution with the satellite antenna pattern. Figures 13a,b show the spatial distribution of polynya pixels detected by the PSSM in the regions around the Ross Ice Shelf and Mertz Glacier Tongue for the same areas as shown in Fig. 6. The Ross Sea coastal polynya, the Terra Nova Bay Polynya, and the Mertz Glacier Polynya are all detected as polynya regions in these figures, shown also in Figs. 6a,b. The shape and location of thin ice (polynya) regions derived from the algorithm of this study (Figs. 6a,b) agree well with those from the PSSM, providing an independent test of the algorithm described in this study. The edge of the Ross Ice Shelf is also well represented by both algorithms. However, some parts of the Mertz and Ninnis Glacier Tongues are misclassified as polynya re-

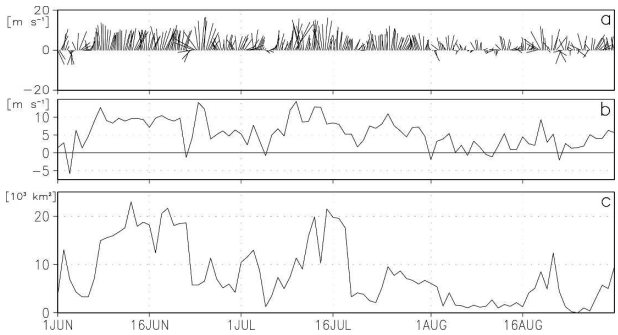

FIG. 12. Time series of (a) stick plots of wind velocity with the offshore component being positive, (b) offshore component of wind velocity, and (c) the total area of thin ice with thickness <0.2 m, for the Ross Sea coastal polynya from 1 Jun to 31 Aug 2001.

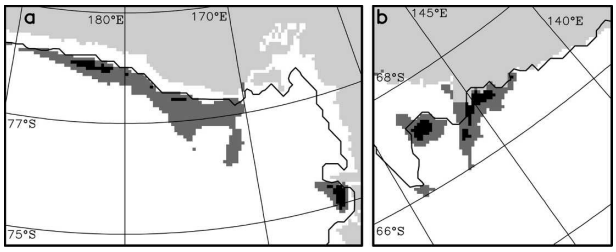

FIG. 13. Spatial distribution of polynyas derived from the PSSM for (a) the region around the Ross Ice Shelf on 10 Aug 1992 and for (b) the region around the Mertz Glacier Tongue on 19 Aug 2000. The black pixels show open water, the dark gray ones show thin ice, and the white ones show thick ice. The solid line indicates the edge of fast ice derived from the algorithm of this study.

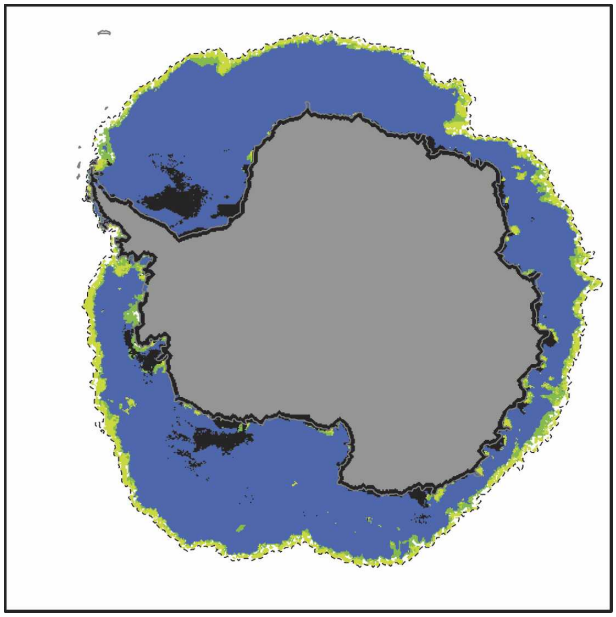

FIG. 14. Spatial distribution of sea ice type derived from the SSM/I algorithm on 9 Jun 1999. The black pixels show fast ice, the blue ones show first-year ice, the green ones show thin ices of 0.1–0.2 m, and the yellow-green ones show thin ices of <0.1 m. The rejected sea ice pixels owing to the atmospheric effects are masked by white. The dashed line outlines the boundary of the 15% sea ice concentration. The gray solid line indicates the coastline of the SSM/I landmask.

gions by the PSSM algorithm, judging from a visual inspection of AVHRR channel-4 imagery (Fig. 3b). The algorithm of this study correctly detects these regions as fast ice and can discriminate thin ice regions nearby. In conclusion, the advantages of the algorithm described here are the estimation of specific thin ice thicknesses and the detection of fast ice regions. The advantages of the PSSM algorithm are its ability to correct for atmospheric effects and its higher spatial resolution.

Figure 14 is an example of the spatial distribution of thin ice thickness and fast ice for 9 June 1999. This figure illustrates that the thin ice region extends all around the Antarctic coasts (green and yellow green pixels). Marginal ice zones are also identified as thin ice regions in this algorithm. At present, we cannot determine whether these regions are really areas of new ice or whether they result from the reduced ice concentration in the marginal ice zones. Microwave characteristics in the marginal ice zone are sensitive to atmospheric effects because of the relatively low latitude (white pixels). With this algorithm, ice shelves, landfast ice, glacier tongues, and large icebergs are clearly detected as fast ice (black pixels). The algorithm tends to regard sea ice areas covered with deep snow as fast ice even for first-year ice. From plate 2 of Markus and Cavalieri (1998), sea ice areas covered with deep snow are found in the Weddell, Ross, and Amundsen Seas. From an Eulerian viewpoint, the center part of these deep-snow areas continues to be deep-snow regions for three months in spite of ice drift. Although pixels with deep snow are erroneously classified as fast ice, this is not a problem for the thin ice algorithm of this study.

Figure 15 shows the relative frequency of thin ice (<0.2 m) occurrence averaged over 1992–2001. It shows that the occurrence of thin ice is generally high around the coast, illustrating the existence of persistent coastal polynyas. Coastal polynyas are frequently formed along the East Antarctic coast, on the west side of peninsulas and glacier tongues, and downstream of the Antarctic Coastal Current. The largest coastal polynya in the Antarctic Ocean is located along the Ross Sea coast. Its size suggests high ice production, which is consistent with the location of known Antarctic Bottom Water (AABW) formation. On the other hand, coastal polynyas occur less frequently along the Weddell Sea coast, suggesting fewer persistent regions of ice production, in spite of it also being known as an AABW formation area. These features were also shown by Arrigo and van Dijken (2003), in which the algorithm of the PSSM was used.

7. Concluding remarks

The ice thickness estimated from the algorithm presented here corresponds to the thickness for which the

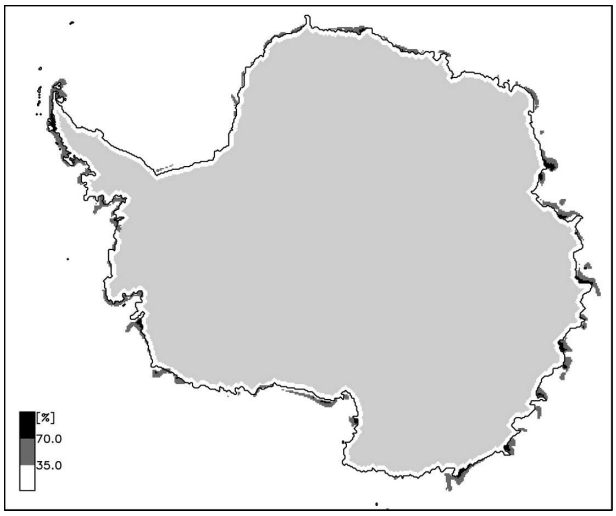

FIG. 15. Relative frequency of thin ice (thickness <0.2 m) occurrence averaged over 1992–2001, derived from the SSM/I algorithm. The black pixels show the area where the frequency is more than 70% and the dark gray pixels show the area where the frequency is 35%–70%. The solid line indicates the coastline of the SSM/I landmask.

calculated total heat loss can be realized under the assumption of a uniform ice thickness in the SSM/I grid cell. It should be noted that the thickness derived from this algorithm does not necessarily correspond to the actual average thickness in the grid cell. Thus we use the term "thermal ice thickness." The retrieved ice thickness is suitable for the estimation of heat flux and the corresponding sea ice production rates in Antarctic coastal polynyas. It should be kept in mind that this algorithm can detect only relatively large polynyas and cannot detect polynyas whose size is less than the SSM/I grid cell scale (10–20 km), such as the relatively narrow coastal polynyas along the Weddell Sea coast.

The thin ice algorithm of this study is based on the relationship between SSM/I PR values and AVHRR thermal ice thicknesses. The AVHRR thermal ice thickness is estimated under clear-sky and freezing conditions. There is no guarantee that this algorithm can be applied to the ice melting season. The relationship between PR values and thin ice thicknesses may also depend on atmospheric conditions. Thus, there is a possibility that the present algorithm has some bias. Also, the discrepancy of the PR–thickness relation between the present Antarctic study and Martin et al.'s (2004) Arctic study is considered to be mainly due to the difference of the atmospheric conditions. Thus, the comparison and validation of ice thickness with the in situ observation under all atmospheric conditions (e.g., using ULS moorings) are required to develop a more general algorithm. On the other hand, ULS data have very different spatial scales compared with SSM/I data. Therefore, AVHRR thermal ice thicknesses will remain the basis for the comparison and validation. The combined comparison and validation with in situ ULS data and AVHRR thermal ice thickness data would be very desirable. We may be able to make the algorithm more robust by adding an atmospheric correction parameter. Given the fact that it is very difficult to obtain in situ ice thickness data that can be compared with the SSM/I data, the present algorithm will serve as the basis for future algorithm improvement.

Acknowledgments. The SSM/I data were provided by the National Snow and Ice Data Center (NSIDC), University of Colorado. The AVHRR data were provided by the National Institute of Polar Research of Japan and NSIDC. We express our appreciation to Seelye Martin for his instructive comments and useful information. We sincerely acknowledge Masaaki Wakatsuchi, Naoto Ebuchi, Toshiyuki Kawamura, Yasushi Fukamachi, Takenobu Toyota, and Noriaki Kimura for their comments and encouragement. We also thank anonymous reviewers for their useful comments. Some of the figures were drawn using the GFD-DENNOU library. T. Tamura was supported by the 21st Century Center of Excellence Program funded by the Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT). K. Ohshima was supported by the fund from the Research Revolution 2002 of the Project for Sustainable Coexistence of Human, Nature and the Earth within MEXT.

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